OPTION PUT–CALL PARITY APPLICATIONS: FIRM VALUE

Learning Outcomes

- explain put–call parity for European options
- explain put–call *forward* parity for European options

The insights established by the put–call parity relationship go well beyond option trading strategies, extending to modeling the value of a firm to describe the interests and financial claims of capital providers—namely, the owners of a firm's equity and the owners of its debt.

Assume that at time $t = 0$, a firm with a market value of V_0 has access to borrowed capital in the form of zero-coupon debt with a face value of D. The market value of the firm's assets, V_0, is equal to the present value of its outstanding debt obligation, PV(D), and equity, E_0: $V_0 = E_0 + \text{PV}(D)$.

When the debt matures at T, the firm's debt and assets are distributed between shareholders and debtholders with two possible outcomes depending on the firm's value at time T (V_T):

1. **Solvency ($V_T > D$)**: If the value of the firm (V_T) *exceeds* the face value of the debt, or $V_T > D$, at time T, we say the firm is *solvent* and able to return capital to *both* its shareholders and debtholders.
   - Debtholders receive D and are repaid in full.
   - Shareholders receive the residual: $E_T = V_T - D$.
2. **Insolvency ($V_T < D$)**: If the value of the firm (V_T) is *below* the face value of the debt, or $V_T < D$, at time T, we say the firm is *insolvent*. In the event of insolvency, shareholders receive nothing and debtholders are owed more than the value of the firm's assets. Debtholders therefore receive V_T to settle their debt claim of D at time T.
   - Debtholders have a priority claim on assets and receive $V_T < D$.
   - Shareholders receive the residual, $E_T = 0$.

Unlike the *risk-free* bond shown in the prior put–call parity lesson, the firm has *risky* debt, because the bondholders receive D *only* in the case of solvency (when $V_T > D$). Debtholders therefore demand a premium similar to a put option premium from shareholders in order to assume the risk of insolvency.

Shareholders retain *unlimited* upside potential (if the firm remains solvent and can pay off its debt) and *limited* downside potential (if the firm becomes insolvent). In contrast, debtholder upside is *limited* to receiving debt repayment in the event of solvency, and principal and interest are at risk in the downside event of insolvency.

Next, we examine the respective payoff profiles at maturity more closely. At $t = T$,

- shareholder payoff is $\max(0, V_T - D)$ and
- debtholder payoff is $\min(V_T, D)$.

Consider these payoff profiles in terms of options:

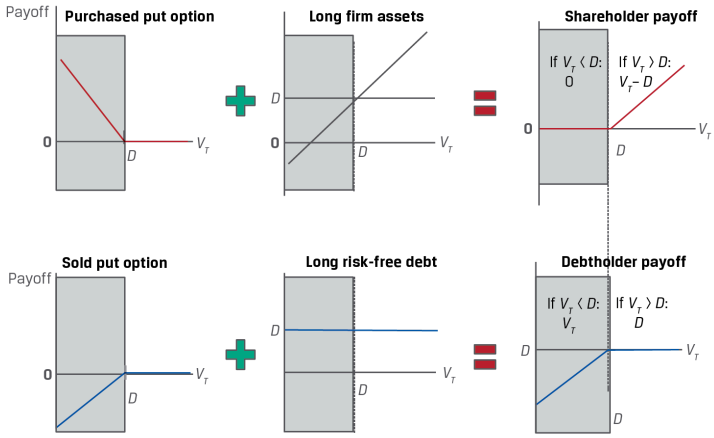
- Shareholders hold a long position in the underlying firm's assets (V_T) and have purchased a put option on firm value (V_T) with an exercise price of D; that is, $\max(0, D - V_T)$.

The payoff profiles for shareholders versus debtholders is shown in Exhibit 9.

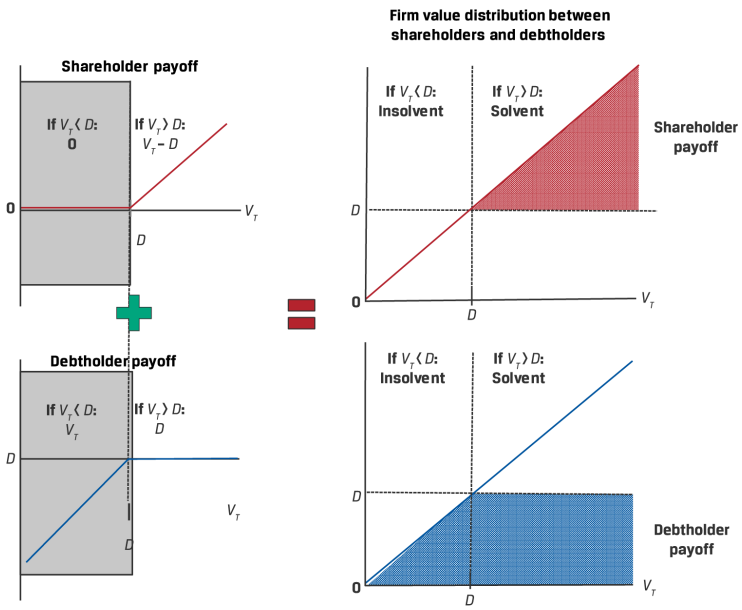
Exhibit 9:

Shareholder and Debtholder Payoff at Time *T*

Construction of Shareholder and Debtholder Payoffs



Firm Value Distribution Between Shareholders and Debtholders



Note that the shareholder's combination of a purchased put option and a long position in the firm's assets is equivalent to a *call* option on the firm's assets. The risky debt held by the debtholders is a combination of the risk-free debt, D, and the put option sold to shareholders.

Revisiting the put–call parity relationship, $S_0 + p_0 = c_0 + \text{PV}(X)$, from Equation 1, we may substitute the value of the firm at time 0 (V_0) for the underlying asset (S_0), substitute the debt (D) for the risk-free bond (X), and solve for V_0 as follows:

$$V_0 + p_0 = c_0 + \text{PV}(D).$$

$$V_0 = c_0 + \text{PV}(D) - p_0. \quad (5)$$

Equation 6 captures the value of the firm's assets at $t = 0$ from a shareholder and debtholder perspective. The *shareholder* has a position with a payoff similar to that of a call option on firm value (c_0). The *debtholder* has a position of $\text{PV}(D) - p_0$, or risk-free debt

pay to debtholders to assume insolvency risk. This put option increases in value to shareholders as the likelihood of insolvency increases. From a debtholder's perspective, the more valuable the sold put, the more credit risk is present in the firm's debt.

Question Set: Put–Call Forward Parity and Option Applications

1. Identify the following statement as true or false, and justify your answer: A key link between put–call parity and put–call forward parity is that the cash underlying position is replaced by a synthetic underlying position using a forward purchase and a risk-free bond.

Hide Solution

Solution:

The statement is true. The replacement of the cash underlying position by a synthetic underlying position using a forward purchase and a risk-free bond links call option, put option, and forward prices under put–call forward parity.

2. Identify which of the following positions has the same no-arbitrage value as which portfolio under put–call forward parity:

1. Long call option (c_0)	A. Forward sale, long risk-free bond, and long call option
2. Long risk-free bond ($X(1 + r)^{-T}$)	B. Forward purchase, long put option, and short risk-free bond
3. Long put option (p_0)	C. Long forward purchase, short call option, and long put option

Hide Solution

Solution:

Recall that the put–call forward parity relationship may be expressed as

$$F_0(T)(1 + r)^{-T} + p_0 = c_0 + X(1 + r)^{-T}.$$

1. B is correct. A long call option position is the no-arbitrage equivalent of a forward purchase, a long put option, and a short risk-free bond position.

2. C is correct. A long risk-free bond position is equivalent to a long forward purchase, a short call option, and a long put option.

3. A is correct. A long put option is equivalent to a forward sale, a long risk-free bond, and a long call option.

3. Describe the cash flows at the time of option expiration for a synthetic protective put.

Hide Solution

Solution:

A synthetic protective put is the combination of a *synthetic* underlying position (using a forward purchase and a long risk-free bond position equal to the exercise price) and a purchased put.

At the time of option expiration, time T, there are two possible scenarios:

returns $S_T - F_0(T)$.

The total return equals $(X - S_T) + [S_T - F_0(T)] + F_0(T) = X$.

Put option unexercised ($X \leq S_T$): If the put option is unexercised, it will expire worthless. The risk-free asset returns $F_0(T)$, and the forward purchase returns $S_T - F_0(T)$.

The total return equals $[S_T - F_0(T)] + F_0(T) = S_T$.

4. Describe how a debtholder's position may be considered similar to the sale of a put option on firm value.

Hide Solution

Solution:

If the value of the firm (V_T) is *below* the face value of its debt outstanding, or $V_T < D$ at time T, we say the firm is *insolvent* and debtholders receive less than the face value (D) to settle their debt claim. Stated differently, a debtholder's payoff is $\min(D, V_T) = D - \max(0, D - V_T)$ and equals the debt face value (D) minus a put option on firm value (V_T) with an exercise price of D, which represents a *sold* put on firm value.

OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

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{OMON} OPTION MONITOR

The Bloomberg OMON function allows you to visualize put–call parity.

> "OMON <GO> provides real-time pricing, market data, and derived data for exchange-traded call and put options all in one customizable screen. You can see options with the greatest volume and open interest, or highlight discrepancies in put-call parity, which enables you to pinpoint arbitrage opportunities. Standard templates provide variations of option data, but you can also create templates tailored to your needs, so you can monitor the market and use the analysis to develop an option strategy. Theoretical pricing and risk column categories allow you to create and manage your own custom volatility surfaces and forwards. Basic scenario analysis is also available in calc mode, providing you the ability to tweak the option model inputs and outputs." - Bloomberg

To explore this function, first type a security—such as Deere & Company (or DE US Equity)—into the Bloomberg command line and press Enter or <GO>. Then select 31) OMON from the popup window. The implied volatility for each call and put listed is in the IVM column. The two tables enable you to highlight discrepancies in put–call parity to spot arbitrage opportunities.

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Calls						Strike	Puts					
Ticker	Bid	Ask	Last	IVM	Volm		Ticker	Bid	Ask	Last	IVM	Volm
16-Jun-23 (4d); CSize 100; R 5.48; IFwd 385.95						5	16-Jun-23 (4d); CSize 100; R 5.48; IFwd 385.95					
1) DE 6/16/23 C380	7.65	8.10	8.05	28.71	100	380.00	46) DE 6/16/23 P380	2.19	2.40	2.33	27.95	116
2) DE 6/16/23 C382.5	6.05	6.40	6.50	28.15	77	382.50	47) DE 6/16/23 P382.5	3.05	3.30	3.10	27.34	132
3) DE 6/16/23 C385	4.65	5.05	4.80	28.21	208	385.00	48) DE 6/16/23 P385	4.10	4.40	3.75	27.59	65
4) DE 6/16/23 C387.5	3.50	3.85	3.72	28.32	159	387.50	49) DE 6/16/23 P387.5	5.40	5.75	5.43	27.53	11
5) DE 6/16/23 C390	2.54	2.81	2.79	28.04	593	390.00	50) DE 6/16/23 P390	6.90	7.35	6.94	27.49	15
21-Jul-23 (39d); CSize 100; IDiv 1.06 USD; R 5.48; IFv						5	21-Jul-23 (39d); CSize 100; IDiv 1.06 USD; R 5.48; IF					
6) DE 7/21/23 C360	29.25	29.90	30.30	25.41	14	360.00	51) DE 7/21/23 P360	3.45	3.65	3.49	25.65	139
7) DE 7/21/23 C370	21.40	22.00	21.65	24.15	4	370.00	52) DE 7/21/23 P370	5.45	5.60	5.50	24.17	36
8) DE 7/21/23 C380	14.55	15.00	14.79	22.82	57	380.00	53) DE 7/21/23 P380	8.60	8.80	8.25	23.05	217
9) DE 7/21/23 C390	9.25	9.40	9.30	21.93	221	390.00	54) DE 7/21/23 P390	13.05	13.40	13.05	22.06	5
10) DE 7/21/23 C400	5.30	5.50	5.55	21.32	193	400.00	55) DE 7/21/23 P400	19.20	19.65	19.83	21.42	102
15-Sep-23 (95d); CSize 100; IDiv 1.06 USD; R 5.49; IF						5	15-Sep-23 (95d); CSize 100; IDiv 1.06 USD; R 5.49; I					
11) DE 9/15/23 C360	38.30	39.10	39.09	27.95	6	360.00	56) DE 9/15/23 P360	9.60	9.80	9.70	27.94	21
12) DE 9/15/23 C370	31.40	31.80	31.92	26.97	17	370.00	57) DE 9/15/23 P370	12.45	12.70	12.45	26.91	9
13) DE 9/15/23 C380	25.00	25.40	25.45	26.11	28	380.00	58) DE 9/15/23 P380	16.00	16.25	15.89	26.07	139
14) DE 9/15/23 C390	19.40	19.75	19.77	25.33	69	390.00	59) DE 9/15/23 P390	20.35	20.65	20.33	25.28	142
15) DE 9/15/23 C400	14.65	14.90	14.85	24.67	59	400.00	60) DE 9/15/23 P400	25.60	25.95	25.70	24.51	3
20-Oct-23 (130d); CSize 100; IDiv 1.94 USD; R 5.53; If						5	20-Oct-23 (130d); CSize 100; IDiv 1.94 USD; R 5.53;					
16) DE 10/20/23 C360	41.70	42.45	39.58y	27.63		360.00	61) DE 10/20/23 P360	11.95	12.45	12.15	27.47	3

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PC

Is there an error in the way pay-off profiles using options have been represented for shareholders and debtholders?

The payoff for a shareholder is max (0, VT - D), which is similar to a call-option on the value of the firms assets with an exercise price of D. What is mentioned as buying a put option and being long on the underlying assets does not create this payoff. If what they have mentioned is considered then during insolvency the put option payoff will be equal to VT + D - VT = D. In case of solvency the payoff will be equal to VT + 0 = VT. Therefore the payoff profile is (D, VT) which is similar to debtholder pay-off? Could someone explain where my assumption is wrong here?

Created 2 months ago by Priyesh Choodamani Iyer — 0 replies | Last Activity: 2 months ago

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NA

Does anyone understand why A debtholder should sell a put option if V < D?

Created 4 months ago by Nada Ait labsir — 1 reply | Last Activity: 4 months ago

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